CLOUD NETWORKING FUNDAMENTALS

Azure Virtual Network

What it is, what it is made of — and how every piece maps to the AWS equivalent you may already know

Facts checked against Microsoft Learn · September 2026

FOUNDATIONS

What a virtual network actually is

A VNet is your own private slice of the Azure network. You choose the IP address range, you decide how it is divided, and nothing outside it can reach in unless you allow it.



Your own address space

You pick the private IP range, for example 10.0.0.0/16. It is yours alone. Two different customers can both use 10.0.0.0/16 and never collide, because the networks are isolated from each other.



Private and isolated by default

Traffic inside a VNet stays on the Microsoft network and never touches the public internet. A resource in one VNet cannot reach a resource in another unless you deliberately connect them.



Scoped to one region, spanning its zones

A VNet lives in a single Azure region. Within that region it stretches across every availability zone — and that one detail is the biggest difference from AWS. Slide 17 returns to it.

FOUNDATIONS

The three things a VNet lets you do

Resources talk to each other

Virtual machines, databases, containers and app services placed in the same VNet can reach each other over private IP addresses. No traffic leaves the Microsoft network.

Resources reach the internet

Outbound connections to public endpoints — patching, calling an API, pulling a container image. Note the word OUTBOUND. Inbound from the internet is a separate, deliberate decision.

Your network reaches your office

A site-to-site VPN over the public internet, or ExpressRoute as a dedicated private circuit. Your data centre and your cloud network behave as one address space.

THE DISTINCTION PEOPLE MISS

Outbound and inbound are completely separate problems. Outbound is about letting your resources reach out. Inbound needs a public IP, a load balancer, an application gateway or a firewall — and each is an explicit thing you create. Nothing in Azure is reachable from the internet by accident.

AND THE MIDDLE ONE IS CHANGING

“Resources reach the internet” used to be free and automatic. For virtual networks created after 31 March 2026 it is not. The next slide is the detail, and it is the single most important slide in the first half.

IMPORTANT CHANGE · VERIFIED SEP 2026

Free outbound internet access has been retired

WHAT CHANGED, AND WHEN

Historically, any VM in Azure could reach the internet outbound with no configuration at all — Azure quietly supplied a shared public IP. Microsoft has retired that behaviour. From 31 March 2026, subnets in NEWLY created virtual networks default to private, and a resource with no explicit outbound method simply cannot reach a public endpoint.

Which network	What happens now	What you must do
Created before 31 Mar 2026	Unchanged — still gets default outbound	Nothing forced, but plan a migration
Created after 31 Mar 2026	Subnets are private by default	Add an explicit outbound method
Any subnet you set private	defaultOutboundAccess = false	Add an explicit outbound method

- **The four explicit methods, best first**
  - NAT Gateway — Microsoft's recommendation for most cases
  - Standard Load Balancer with outbound rules
  - A Standard public IP on the NIC (fine for one VM, not a pattern)
  - Firewall or NVA reached by a user-defined route

WHY THIS IS GOOD NEWS

Silent, shared, un-auditable outbound access was always a weak default — you could not see it, control it or attribute it. An explicit NAT Gateway gives you a predictable source IP that a partner can allow-list, and a place to observe and control what leaves.

CORE COMPONENTS

Subnets — dividing the address space

A subnet is a slice of your VNet's address range. You put resources into subnets, not directly into the VNet, and most controls — security rules, routing, service access — attach at the subnet level.

A TYPICAL LAYOUT

```
VNet         10.0.0.0/16      65,536 addresses

  snet-web     10.0.1.0/24     251 usable
  snet-app     10.0.2.0/24     251 usable
  snet-data    10.0.3.0/24     251 usable
  snet-pep     10.0.4.0/24     251 usable
```

- **Rules worth knowing**
  - Azure reserves 5 addresses in every subnet
  - Smallest usable subnet is /29; /24 is the common default
  - Subnets cannot overlap within a VNet
  - Some services demand a dedicated subnet of their own
  - Resizing a subnet with resources in it is painful

THE FIVE RESERVED ADDRESSES

In 10.0.1.0/24 you lose .0 (network), .1 (gateway), .2 and .3 (Azure DNS), and .255 (broadcast). So 256 becomes 251.

PLAN BIGGER THAN YOU NEED

Address space is free. Growing a subnet later, once resources sit in it, is not. Start at /24 even for three servers.

A PRACTICAL NAMING HABIT

Name subnets by role, not by number — snet-app-prod rather than subnet-2. Six months later, when somebody is reading a network security group rule at 2am, the name is the only context they have about what the rule is protecting.

CORE COMPONENTS

Network Security Groups — the firewall rules

An NSG is a list of allow and deny rules for traffic in and out. You attach it to a subnet, to a single network interface, or to both. It is stateful: allow a connection in and the reply is allowed out automatically.

Priority	Rule	Direction	Effect
100	Allow 443 from Internet	Inbound	Your public web traffic
200	Allow 1433 from snet-app	Inbound	App tier reaches the database
4096	DenyAllInBound (built in)	Inbound	Everything else is refused
65000	AllowVnetInBound (built in)	Inbound	Anything inside the VNet is allowed
65001	AllowAzureLoadBalancer (built in)	Inbound	Health probes work

LOWEST NUMBER WINS

Rules are evaluated by priority, lowest first, and the first match stops evaluation. A permissive rule at priority 100 silently defeats a strict rule at 200. When a rule “does nothing”, look above it.

THE DEFAULT THAT SURPRISES PEOPLE

AllowVnetInBound means anything already inside your VNet can reach anything else by default. The VNet is closed from outside and open on the inside. Segmenting the inside is a decision you must make deliberately.

CORE COMPONENTS

VNet peering — joining two networks

Peering connects two VNets so their resources reach each other by private IP, as if they were one network. Traffic travels on the Microsoft backbone — never the public internet, no gateway, no VPN, no encryption overhead.

Same region or different	Regional peering within a region, global peering across regions. Both work the same way.
Address spaces must not overlap	This is the number one reason a peering fails. Plan the address ranges before anybody deploys.
Not transitive	If A peers with B, and B peers with C, then A cannot reach C. You must peer A and C directly, or route through an appliance in B.
Both sides must agree	A peering has two halves. Create one and the link stays Initiated until the other side is created too.
You pay for data moved	Ingress and egress are charged. Cross-region peering costs more than peering inside a region.

CORE COMPONENTS

Routing — what Azure does for you

You do not have to create routing for a VNet to work. Azure gives every subnet a set of system routes automatically. You only write routes when you want to override that default behaviour.

Route	Where traffic goes	Created by
Your VNet address space	Stays inside the VNet	Azure, automatically
Peered VNet ranges	Across the peering link	Azure, when you peer
On-premises ranges	To the VPN or ExpressRoute gateway	Azure, when you connect
0.0.0.0/0 (everything else)	Out to the internet	Azure, automatically
Your override	Wherever you send it	You — a user-defined route

THE MOST COMMON OVERRIDE

Send 0.0.0.0/0 to a firewall instead of straight out. Now every outbound packet is inspected and logged before it leaves. This one route is the backbone of most enterprise designs.

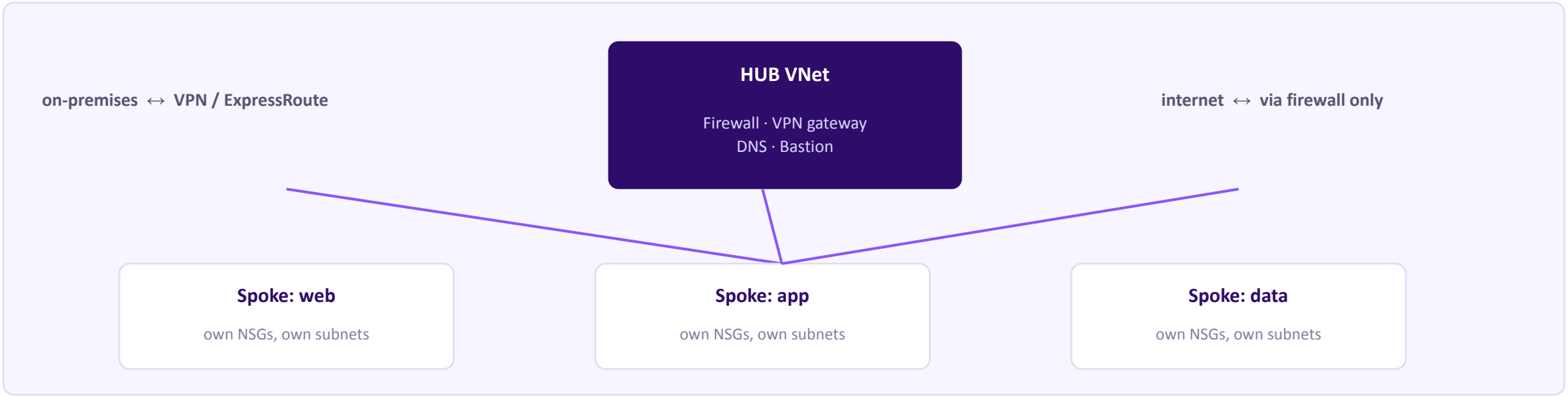
ROUTES AND NSGS ARE DIFFERENT THINGS

A route decides WHERE a packet goes. An NSG decides WHETHER it is allowed. Both can break a connection, and the symptoms look identical from the application.

FOUNDATIONS

Putting it together — a shape you will see everywhere

Hub and spoke. One central VNet holds shared services and the security controls; each workload gets its own VNet peered to the hub. It exists because peering is not transitive.



WHY THIS SHAPE WINS

One place to inspect traffic, one VPN gateway instead of five, one DNS story, and a blast radius per workload. Spoke-to-spoke traffic is forced through the hub firewall, which turns non-transitive peering from a limitation into a control. The AWS equivalent is a Transit Gateway with attached VPCs — same idea, different machinery.

AZURE ↔ AWS

The translation table — start here

Most concepts map cleanly. Learn these names and eighty per cent of an AWS networking conversation becomes readable in Azure, and the reverse.

Feature	Azure	AWS	What it means for you
Private network	Virtual Network (VNet)	Virtual Private Cloud (VPC)	Same idea
Network slice	Subnet	Subnet	Different scope — see slide 17
Instance firewall	Network Security Group	Security Group	NSG also works on subnets
Subnet firewall	Network Security Group	Network ACL	AWS NACL is stateless
Tag-based grouping	Application Security Group	SG referencing an SG	Same purpose
Network joining	VNet peering	VPC peering	Both non-transitive
Managed hub	Virtual WAN	Transit Gateway	Both solve the same problem
Custom routing	User-defined route	Route table entry	Azure has system routes
Outbound NAT	NAT Gateway	NAT Gateway	Now required in both
Dedicated line	ExpressRoute	Direct Connect	Same idea

Network fundamentals, compared

Feature	Azure	AWS	What it means for you
Scope of network	One region	One region	Same
Scope of subnet	Regional — spans all zones	Zonal — one zone only	MAJOR difference
Zone redundancy	One subnet is enough	One subnet per zone	Affects every design
Reserved IPs per subnet	5	5	Same count
Smallest subnet	/29	/28	Minor
Address space	Multiple ranges, can add	Primary + secondary CIDRs	Similar
“Public subnet” concept	Does not exist	Defined by route to IGW	Different model
Default route table	System routes, automatic	Main route table	Similar
DNS	Azure-provided at x.x.x.2/.3	AmazonProvidedDNS at .2	Similar

Row 2 is the one to remember. In AWS a subnet lives in exactly one availability zone. In Azure a subnet spans them all. Every high-availability design in each cloud follows from that single fact.

Security controls, compared

Feature	Azure	AWS	What it means for you
Instance-level firewall	NSG on the NIC	Security Group on the ENI	Similar
Subnet-level firewall	The same NSG	Network ACL	Different design
Stateful?	Yes, always	SG yes / NACL no	Watch the NACL
Deny rules	Yes	SG no / NACL yes	Different
Default inbound	Deny from outside	Deny all	Similar
Default inside network	ALLOWED between resources	Denied unless SG allows	MAJOR difference
Rule ordering	Priority number, first match	SG any-match / NACL by number	Different
Logical grouping	Application Security Group	SG referencing an SG	Same purpose
Managed firewall	Azure Firewall	Network Firewall	Similar
Traffic logging	NSG flow logs	VPC Flow Logs	Similar

Row 6 is the trap in both directions. Azure allows traffic between resources in a VNet by default; AWS denies it until a security group permits it. Carrying either assumption across clouds produces a real security gap.

Connectivity and hybrid, compared

Feature	Azure	AWS	What it means for you
Network to network	VNet peering	VPC peering	Both non-transitive
Across regions	Global VNet peering	Inter-region VPC peering	Similar
Many networks at scale	Virtual WAN	Transit Gateway / Cloud WAN	Similar
Share a VPN gateway	Gateway transit over peering	Not available on peering	Azure advantage
Encrypted tunnel to office	VPN Gateway	Site-to-Site VPN	Similar
Dedicated private circuit	ExpressRoute	Direct Connect	Similar
Individual remote user	Azure VPN Client / P2S	AWS Client VPN	Similar
Outbound internet	NAT Gateway (now required)	NAT Gateway + IGW	See slide 19
Inbound internet	Public IP, LB, App Gateway	IGW + ELB	Different model
Admin access to a VM	Azure Bastion	Session Manager / EIC	Similar intent

Row 4 is a genuine Azure advantage. A spoke can use the hub's VPN or ExpressRoute gateway over the peering link — so one expensive circuit serves every workload. AWS peering cannot share a gateway; that is a reason people adopt Transit Gateway.

Reaching managed services privately, compared

Both clouds let you reach a managed service — storage, a database, a model endpoint — without the traffic crossing the public internet. The machinery differs more here than anywhere else.

Feature	Azure	AWS	What it means for you
Private IP for a service	Private Endpoint	Interface VPC Endpoint	Same idea
Technology name	Azure Private Link	AWS PrivateLink	Same idea
Route-based shortcut	Service Endpoint	Gateway VPC Endpoint	Azure is broader
Does it give a private IP?	Private Endpoint yes	Interface endpoint yes	Same
Reachable from on-prem	Yes, via private endpoint	Yes, via interface endpoint	Same
Expose your own service	Private Link Service	Endpoint Service	Same
Cost shape	Per endpoint + data	Per endpoint + data	Similar

PRIVATE ENDPOINT VS SERVICE ENDPOINT

A private endpoint gives the service a real private IP inside your subnet. A service endpoint just keeps traffic on the Azure backbone. The endpoint is stronger and works from on-premises.

THE DNS TRAP

Private endpoints need private DNS zones to resolve correctly. Miss this and the name still resolves — to the public IP. It looks like it works, and the traffic goes the wrong way.

CROSSING OVER

Five differences that actually cause outages

Most of the mapping is safe. These five are where a correct AWS habit produces a wrong Azure result, and the reverse.

1	**Subnets are regional, not zonal**	You do not create one subnet per zone. You choose zones when you place the resource.
2	**One stateful NSG, not a security group plus a stateless ACL (see slide 12)**	Azure has no stateless layer. AWS NACLs need both directions written out.
3	**No internet gateway, and no “public subnet”**	Azure has no gateway object to attach. Stop looking for the setting.
4	**Peering can share a gateway**	Gateway transit lets a spoke use the hub's VPN or ExpressRoute. VPC peering cannot.
5	**Inside the network is open by default**	Azure allows VNet-internal traffic unless you stop it. AWS denies until you allow.

DIFFERENCE 1

Subnets: zonal in AWS, regional in Azure

AWS

A subnet belongs to exactly ONE availability zone.

subnet-a → ap-south-1a
subnet-b → ap-south-1b
subnet-c → ap-south-1c

Want three-zone redundancy? Create three subnets, and your load balancer spans them.

AZURE

A subnet spans EVERY zone in the region.

snet-app → the whole region

Zone is chosen when you create the VM, not when you create the subnet. Three VMs in three zones can share one subnet and one address range.

WHAT GOES WRONG

An AWS engineer arrives in Azure and builds snet-app-1, snet-app-2 and snet-app-3, then cannot find where to set the zone. There is nowhere — the subnet has no zone. They end up with three subnets doing the job of one, three sets of NSG rules to keep in step, and a mistaken belief that they have zonal isolation.

THE RULE TO CARRY

In AWS the subnet is the zone decision. In Azure the resource is the zone decision. So in Azure, split subnets by ROLE — web, app, data, private endpoints — never by zone. Then set the zone on each virtual machine, or use a zone-redundant service that handles it for you.

DIFFERENCE 3 · DIFFERENCE 2 IS ON SLIDE 12

There is no internet gateway in Azure

In AWS, a VPC reaches the internet only when you create an internet gateway, attach it, and add a route. “Public subnet” simply describes a subnet whose route table points at that gateway. Azure has no such object.

To do this	In AWS	In Azure
Outbound to internet	IGW + route + public IP or NAT GW	NAT Gateway (now required on new VNets)
Inbound to a VM	IGW + public IP + SG rule	Public IP on the NIC + NSG rule
Inbound to a service	IGW + load balancer	Load Balancer, App Gateway or Front Door
Make a subnet private	Point its route table away from IGW	It already is — that is the default

WHY THE DIFFERENCE EXISTS

AWS makes connectivity a property of the NETWORK — attach a gateway, add a route. Azure makes it a property of the RESOURCE — give it a public IP, or a NAT Gateway on its subnet.

WHAT PEOPLE SEARCH FOR AND NEVER FIND

“How do I make this subnet public in Azure?” There is no such setting. The question itself is an AWS-shaped question.

The two models converge in practice. Since March 2026 Azure also requires an explicit NAT Gateway on new VNets — so both clouds now make outbound internet access a deliberate, visible, auditable decision rather than a silent default.

DIFFERENCE 4

Peering that can share a gateway

Both clouds have non-transitive peering. Azure adds one capability on top of it that meaningfully changes hub-and-spoke economics.

- **Two settings on an Azure peering**
  - Allow gateway transit — set on the HUB side
  - Use remote gateways — set on the SPOKE side
  -
- **What it buys you**
  - One VPN or ExpressRoute circuit for the whole estate
  - A new spoke inherits hybrid connectivity by peering alone
  - No gateway to deploy, pay for or maintain per workload

IN AWS

VPC peering cannot share a gateway. Each VPC needing hybrid connectivity either gets its own Virtual Private Gateway, or you adopt Transit Gateway — which is the usual answer, and is also a managed service with its own cost.

STILL NOT TRANSITIVE

Gateway transit shares the GATEWAY. It does not make peering transitive. Spoke A still cannot reach Spoke C through the hub unless you route that traffic through a firewall or appliance sitting in the hub.

Practical consequence: in Azure, hub-and-spoke with gateway transit is often enough. In AWS you reach for Transit Gateway sooner, because peering alone will not carry hybrid connectivity to the spokes.

DIFFERENCE 5

The defaults, side by side

Feature	Azure	AWS	What it means for you
Traffic inside the network	ALLOWED by default	Denied until an SG allows	Azure is open inside
Traffic from the internet	Denied	Denied	Same
Outbound to internet	Explicit method needed (new VNets)	IGW + route needed	Now similar
Reply traffic	Automatic — stateful	SG yes / NACL no	Watch the NACL
Load balancer probes	Allowed by a built-in rule	SG rule required	Azure is easier
DNS resolution	Automatic inside the VNet	Automatic inside the VPC	Same

ROW 1, IN BOTH DIRECTIONS

Moving to Azure: assuming denied-by-default gives you a flat network where one compromised machine reaches everything. Moving to AWS: assuming allowed-by-default gives you a morning lost to two servers that cannot see each other.

The fix in Azure is not complicated: an NSG per subnet with explicit rules, or Application Security Groups so rules read as roles rather than address ranges.

APPLYING IT

What to decide before anybody deploys

Address plan	Pick ranges that will never overlap with another VNet, an office, or an acquisition. Peering fails permanently if they do, and renumbering later is a project.
Subnet layout by role	web, app, data, private endpoints, and a dedicated subnet each for Firewall, Bastion and any gateway. Size generously — address space is free, resizing is not.
Outbound method	Not optional on new VNets since March 2026. Choose NAT Gateway unless you have a reason not to, and decide whether traffic must pass a firewall first.
Segmentation inside	Azure allows VNet-internal traffic by default. Decide what should be able to reach the data tier, and write it down as NSG or ASG rules.
Hybrid connectivity	VPN or ExpressRoute, in the hub, with gateway transit on so spokes inherit it. One circuit, not one per workload.
Private access to PaaS	Private endpoints for anything holding real data — and the private DNS zones to go with them. Verify resolution from inside the network.

APPLYING IT

Mistakes worth avoiding once rather than twice

Overlapping address ranges	Two VNets both on 10.0.0.0/16 can never be peered. Discovered at the worst moment, usually during a merger or when connecting to a partner.
Subnets too small	A /27 looked generous for six servers. Private endpoints, a scale-out event and a platform upgrade later, it is full, and growing it means moving everything.
Forgetting the outbound method	On a VNet created after March 2026 the VM cannot patch, cannot pull an image, cannot call an API. The error looks like DNS or a firewall, not a missing NAT Gateway.
Trusting the inside of the VNet	Default rules let every resource reach every other. One compromised web server then reaches the database directly. Segment deliberately.
Private endpoint without private DNS	The name still resolves — to the public IP. Everything appears to work while the traffic takes the path you were trying to avoid.
Expecting peering to be transitive	A to B and B to C does not give A to C. Route through an appliance in the hub, or peer directly.
A permissive NSG rule at a low priority	First match wins, so an allow rule at 100 silently defeats every stricter rule below it. When a rule seems inert, look above it.

REFERENCE

The words, in plain language

VNet

Your private, isolated network in one Azure region.

Subnet

A slice of the VNet's address range. Controls attach here.

NSG

Stateful allow/deny rules, on a subnet or a network card.

ASG

A label for a group of machines, so rules read as roles.

Peering

A direct private link between two VNets. Not transitive.

Gateway transit

Letting a peered spoke use the hub's VPN or ExpressRoute.

UDR

A route you write to override Azure's automatic routing.

NAT Gateway

The managed way out to the internet. Now required on new VNets.

Private Endpoint

A private IP inside your subnet for a managed service.

Service Endpoint

Keeps traffic to a service on the Azure backbone. Weaker.

ExpressRoute

A dedicated private circuit to your data centre.

Hub and spoke

Shared services in the middle, one VNet per workload around it.

SUMMARY

Five things to carry away

A VNet is a private address space you own, closed from outside and open on the inside.

The second half of that sentence is the one people forget.

Subnets are regional in Azure, zonal in AWS.

Split subnets by role, then choose zones on the resource.

One stateful NSG does the work of a security group and a network ACL.

And AWS network ACLs are stateless, so both directions must be written.

Free outbound internet access ended on 31 March 2026 for new VNets.

Choose a NAT Gateway deliberately. Both clouds now want an explicit one.

Peering is not transitive — but in Azure it can share a gateway.

That is why hub-and-spoke with gateway transit is often enough.

Checking your own facts

Cloud networking defaults change, and a lot of published material still describes the old outbound behaviour as current.

Before you quote a default in front of a client, check the vendor's own documentation page for that setting.

Everything in this deck was verified in September 2026.

Sources: Microsoft Learn — Virtual Network overview, Default outbound access, and the AWS-to-Azure networking comparison.